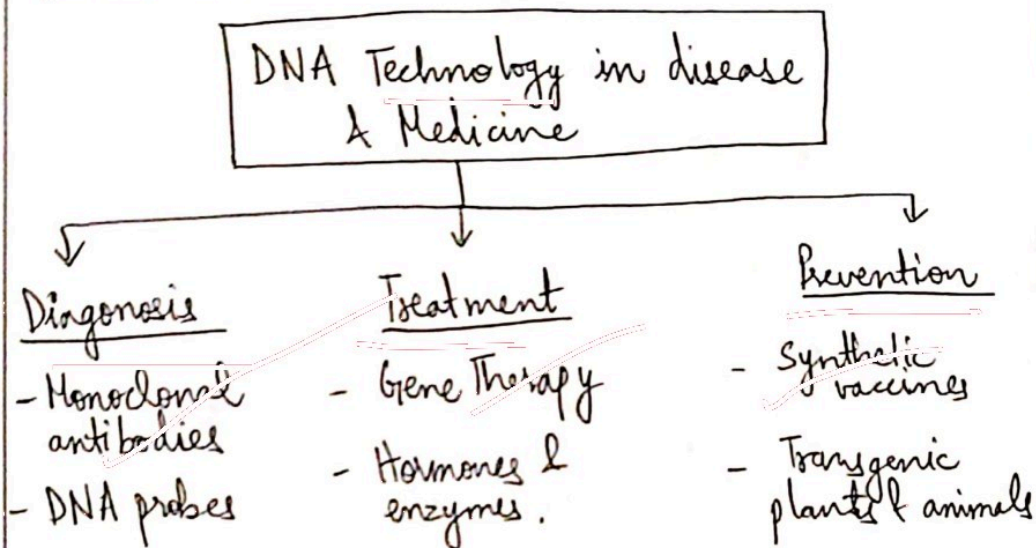


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this margin(c) Critically examine the role of DNA technology in diseases and medicine 15

DNA technology refers to the alteration & manipulation of DNA to get desired outcomes.



I. Diagnosis

(i.) Monoclonal antibodies

- These are antibodies which only sticks to a single antigen.
- They are created ~~using~~ through ~~lymphocytes~~ leukocytes ~~then~~ using ~~diapase~~ hybridoma technique.

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ii) DNA probes

- They are the DNA segments that attach to the complementary bases of other DNA
- Many parasitic ~~insects~~ like plasmodium can be identified.

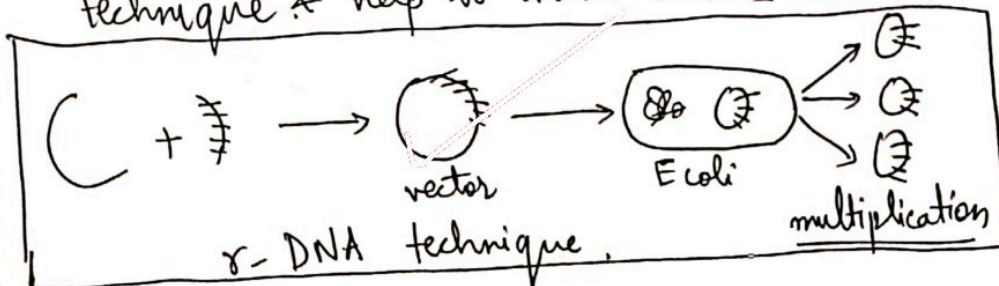
II. Treatment

i) Gene ~~Therapy~~ Therapy

- involves 3 processes
 - gene augmentation
 - gene replacement
 - gene transfer.
- Various disease like ADA deficiency are treated.

ii) Hormones & enzymes

- Various hormones are created through r-DNA technique & help to treat various diseases.



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III. Prevention

(i) Synthetic vaccines

- Unlike conventional vaccines, they do not use germs to create vaccines, hence relatively less harmful.
- used to treat diseases like rabies, cholera etc.

Application in
Medical
jurisprudence

(ii) Transgenic plants & animals

- various crops like GM mustard are designed to withstand pest attack.
- Various disease susceptible animals are also produced through crossing.

Hence, DNA technology has huge application in disease & medicine. New technologies like, CRISPR-CAS 9, HGP-WIRE are promising.

The directive of the question is Critically examine

You need to mention limitations of DNA technology

Mention about ethical issues involved

Example

Three parent baby

Designer babies

Stem cell therapy (Destroying embryo for embryo cells)